Lecture 4.2: Writing the Conceptual Framework

AREC 570 / ECON 530 · Unit 4

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Readings and sequence

1. **Chouinard, “Theory Notes”**
    - What belongs in the theory section of an empirical paper
2. **Cheng, “How to Write Proofs: A Quick Guide”**
    - How to make each step in an argument follow from what precedes it

Presenting the model in a paper

Lecture 4.1 developed a conceptual model of retailer authorization.

The next step is to present that reasoning in a paper

- Explain why the theory is relevant to the research problem
- Define the agents, choices, program rules, assumptions, and relationships
- Show how the prediction follows from the model
- Connect the prediction to the available evidence

Recall the retailer participation model

For retailer i

- B_i is the benefit from SNAP authorization including SNAP-related sales
- $K_i(r)$ is the cost of meeting stocking standard r
- $a_i(r)$ records whether the retailer is authorized

$$a_i(r) = 1 \quad \text{when} \quad B_i \geq K_i(r)$$

A stronger standard may raise $K_i(r)$ by different amounts across retailers. If the cost rises above B_i the model predicts a change from authorized to unauthorized.

Review the following claim

> Because stricter stocking standards require healthier foods, they improve food access and diet quality.

Use the retailer model on the previous slide to assess the claim.

What does the model predict?

The standard may raise compliance costs and cause some retailers to lose authorization which can reduce the number of SNAP outlets in a community

What do the available data observe?

Whether stores remain authorized and how the authorized-store network changes

What requires additional evidence?

Changes in inventory food availability purchases diet and health

Chouinard: Structuring theory

Theory in an empirical paper

Chouinard emphasizes making the theory that motivates the research question and hypotheses explicit (Chouinard 2015).

For an empirical paper the theory section can

- Explain the mechanism behind the research question
- State the assumptions needed for that mechanism
- Derive the hypotheses evaluated with data
- Connect theoretical concepts to empirical measures
- State the limits of the conclusions supported by the evidence

The role of theory depends on the contribution

Theoretical paper	Empirical paper
Develop or modify a model	Apply theory to explain the empirical question
Establish analytical results	Derive hypotheses and comparative predictions
Show what changes relative to prior theory	Connect concepts to observed measures
Treat the model as a central contribution	Use the model to organize the empirical argument

Both require explicit assumptions and a coherent sequence of reasoning.

Varying degrees of theory in a paper

Apply established theory

Summarize the relevant result and explain how it applies to the research problem

Modify established theory

State the baseline model, identify the change, and show how the result changes

Develop new theory

Define the setting, assumptions, relationships, and results in sufficient detail to establish the contribution

The appropriate level of detail depends on how theory contributes to the paper.

Present the model in a logical order

Chouinard recommends introducing the model components before presenting the complete argument.

1. Define the setting and relevant agents
2. Identify choices and institutional constraints
3. State the maintained assumptions
4. Introduce the relationships among model objects
5. Derive and interpret the conditional result
6. Connect the result to the empirical hypothesis

Apply the sequence: setting, agents, choices, and constraints

Setting and agents SNAP-authorized retailers respond to a revised stocking standard

Choice Each retailer decides whether authorization remains worthwhile

Institutional constraint A retailer must satisfy the stocking standard to remain eligible for SNAP authorization

The stocking rule limits the retailer’s feasible choices.

A compliance cost is a consequence of responding to that rule rather than the constraint itself.

Apply the sequence: assumptions, relationship, and result

Maintained assumptions

These are conditions accepted for the model comparison

- The stronger standard does not reduce compliance cost and may raise it by different amounts across retailers
- B_i is held fixed when comparing the two standards
- Retailers compare the benefit of authorization with the cost of compliance

Relationship $a_i(r) = 1$ when $B_i \geq K_i(r)$

Conditional result Authorization changes when $K_i(r)$ rises from no greater than B_i to greater than B_i

Interpret the central terms consistently

For retailer i

- r is the strictness of the stocking standard
- B_i is the benefit of authorization including SNAP-related sales
- $K_i(r)$ is the cost of meeting standard r
- $a_i(r)$ is authorization status

The index i allows benefits and costs to differ across retailers.

Pair the equation with its interpretation

The retailer remains authorized when the benefit of authorization covers the compliance cost

$$a_i(r) = \begin{cases} 1 & \text{if } B_i \geq K_i(r) \\ 0 & \text{if } B_i < K_i(r) \end{cases}$$

A change in the standard changes authorization only when compliance cost moves from at or below B_i to above B_i.

Separate the model result from the empirical hypothesis

Model result

Holding B_i fixed, a higher $K_i(r)$ can cause a retailer to lose authorization

Empirical hypothesis

Retailers expected to face higher compliance costs are more likely to lose authorization after implementation

Evidence

Store-year authorization data can show whether losses differ by observable retailer characteristics

The evidence may support or challenge the hypothesis without revealing the unobserved compliance cost itself.

Discussion: apply Chouinard to your research

1. What role does theory serve in your paper
2. Are you applying modifying or developing a model
3. Which concept must be defined before the reader sees an equation
4. Which assumption carries your principal result
5. What empirical hypothesis follows from that result

Cheng: Explicit reasoning

Cheng's structure: premises reasoning and conclusion

Cheng describes a proof as a sequence of statements in which each statement follows from what came before (Cheng 2004).

Part	Function in an economic argument
Beginning	Definitions assumptions and accepted results
Middle	The steps connecting the premises to the result
End	The claim the argument was constructed to establish

The written argument makes the intermediate steps visible.

Develop the argument backward and present it forward

When developing an argument it is useful to ask

> What would have to be true for this conclusion to follow?

That question works backward from the intended result and helps identify the required assumptions and intermediate steps.

The written argument should then proceed forward from the stated premises to the result.

Apply Cheng’s structure to the retailer model

Beginning

The revised standard may raise $K_i(r)$ while B_i is held fixed

Middle

A retailer remains authorized only when $B_i \geq K_i(r)$

End

For a retailer whose compliance cost rises above the benefit of authorization the model predicts a change from authorized to unauthorized

Show the threshold crossing

Suppose the standard changes from r_0 to r_1 and

$$K_i(r_0) \leq B_i < K_i(r_1)$$

Then

$$a_i(r_0) = 1 \quad \text{and} \quad a_i(r_1) = 0$$

The retailer was authorized under the earlier standard and is not authorized under the stronger standard.

“Handwaving”

Handwaving is a term that describes a step that may be plausible but is not established by the stated argument

The authorization model can support

$$\text{higher compliance cost} \rightarrow \text{possible authorization loss}$$

It does not by itself establish

$$\text{authorization loss} \rightarrow \text{food access change} \rightarrow \text{diet or health change}$$

That downstream pathway requires relationships and evidence about inventory availability purchases and health.

Two claims that require more support

Invalid converse

> A retailer remained authorized therefore the revised requirement was slack

The retailer may instead have changed its inventory to comply.

Unsupported causal attribution

> A retailer lost authorization after implementation therefore the revised rule caused the loss

The analysis needs to distinguish policy exposure from closure, reclassification, and other changes over time.

Discussion: trace a claim in your research

Choose one claim your conceptual framework is intended to support

1. What conclusion follows from the model
2. Which maintained assumption is needed for that conclusion
3. What observed result would be consistent with the conclusion
4. What stronger claim would require another relationship or additional evidence

Synthesis

Organizing a written conceptual framework

1. State the research problem that requires explanation
2. Identify the relevant economic theory
3. Define agents, choices, institutional constraints, and key terms
4. State the maintained assumptions
5. Introduce the relationships among model objects
6. Derive the result and empirical hypothesis
7. State what the model and available evidence do not establish

The sequence should make the relationship between theory, prediction, and evidence clear.

Use the framework to inform the empirical design

Step	Question	SNAP application
Framework	What mechanism produces the prediction?	A stronger standard may change authorization through compliance cost
Measurement	Which variables represent the concepts?	Authorization, retailer type, location, and local SNAP participation
Identification	What comparison isolates the policy response?	A comparison to what authorization would have been without the change

SNAP example: problem and structure

Research problem

Revised stocking standards may change the number, type, and location of SNAP-authorized retailers

Economic theory

Retailer participation is a threshold choice in which the benefit of authorization is compared with the cost of compliance

Agent and choice

Retailer i decides whether the benefit of authorization justifies the cost of meeting the standard

Institutional constraint

The retailer must satisfy stocking standard r to remain authorized

SNAP example: assumptions and result

Maintained assumptions

- A stronger standard does not reduce $K_i(r)$ and may raise it by different amounts across retailers
- B_i is held fixed when comparing standards
- Retailers compare the benefit of authorization with the cost of compliance

Relationship $a_i(r) = 1$ when $B_i \geq K_i(r)$

Conditional result

$$K_i(r_0) \leq B_i < K_i(r_1) \quad \Rightarrow \quad a_i(r_0) = 1 \text{ and } a_i(r_1) = 0$$

SNAP example: hypothesis, evidence, and audit

Empirical hypothesis

Authorization losses increase after implementation among retailers expected to face larger compliance-cost increases

Observable evidence

Store-year data show authorization changes by retailer type, location, and local SNAP participation

Takeaways

The argument establishes a conditional prediction about authorization

Changes in inventory, household food access, purchases, diet, or health require additional relationships and evidence

SNAP example: in paragraph form

> The revised standard may raise the cost of continued SNAP authorization, with the increase varying across retailers. Let B_i denote the benefit a retailer receives from authorization, including SNAP-related sales, and let $K_i(r)$ denote the compliance cost. Holding B_i fixed, the model predicts an authorization loss when $K_i(r)$ exceeds B_i. This prediction motivates an analysis of authorization changes by store type and local SNAP participation. The available data do not establish changes in inventory, household access, purchases, diet, or health.

SNAP example: paragraph checklist

- States the maintained assumptions
- Separates model results from empirical hypotheses
- Distinguishes observed from unobserved outcomes
- Limits the conclusion to what the evidence can support

SNAP example: boundaries on the scope of the evidence

The framework supports a conditional prediction about retailer authorization.

The data can describe changes in the number, type, and location of authorized retailers.

The study cannot directly establish

- Which retailers changed inventory and by how much
- How household access to qualifying food changed
- How household purchases changed
- Whether diet or health improved

Workshopping conceptual frameworks

In your groups: build a framework

Choose one volunteer whose research question the group will workshop

1. State the research problem and outcome
2. Identify the relevant economic theory
3. Define the agents, choices, and institutional constraints
4. State the maintained assumptions and relationships
5. Derive a conditional prediction and empirical hypothesis

15 minutes · groups of 3–4

As a class: present and audit

Group report

- The research problem and conditional prediction
- One empirical hypothesis
- One step requiring an additional premise or evidence

Class audit

- Check whether each step follows from the preceding step
- Identify a missing premise, unsupported inference, or outcome outside the evidence

Output: A preliminary framework that connects the premises to the conditional prediction

Apply the reporting sequence to your research

For your paper, ask

1. What mechanism connects the cause or policy to the outcome?
2. Which maintained assumptions are required for the prediction?
3. Which theoretical concepts have observable counterparts?
4. What empirical comparison would isolate the proposed effect?
5. Which conclusions remain outside the model or available evidence?

Summary of readings

Chouinard identifies the components and order of a theory section

Cheng requires each conclusion to follow from stated premises

Both provide guidelines to turn a model into a written argument whose scope a reader can evaluate.

References

Cheng, Eugenia. 2004. *How to Write Proofs: A Quick Guide*. Department of Mathematics, University of Chicago.
Chouinard, Hayley H. 2015. *Theory Notes*. EconS 529 course notes.

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